

DET-004 | PowerShell Encoded Command Execution

ID: DET-004

MITRE ID: T1059.001

Data Source: DeviceProcessEvents

Severity: High

Tactic: Execution

Category: Endpoint

ADX MAPPING LOGIC:

Mapping: EpisodeId = 'ProcessId', EventType = 'ProcessName', DeathDirect (bool) = 'EncodedFlag', State = 'DeviceName'. We use extract() and has_any() to simulate regex-based command-line scanning.

OBJECTIVE: Practice extract(), has_any(), and regex matching — the core operators used to detect Base64-encoded PowerShell in DeviceProcessEvents. In real Sentinel this filters ProcessCommandLine for -EncodedCommand / -enc flags.

KQL QUERY (RUNNABLE ON ADX FREE TIER)

```
StormEvents
| where StartTime > ago(365d)
| where EventType has_any (
"Thunderstorm", "Tornado", "Flash Flood" // simulates suspicious process names
)
| extend
ExtractedCode = extract(@"([A-Z][a-z]+\s[A-Z][a-z]+)", 1, EpisodeId),
SuspiciousFlag = iff(
DeathDirect == true, "ENCODED_FLAG_SET", "NORMAL"
)
| where SuspiciousFlag == "ENCODED_FLAG_SET"
| project
StartTime,
DeviceName = State,
ProcessName = EventType,
EpisodeId,
SuspiciousFlag,
ExtractedCode,
Source
| order by StartTime desc
| take 20
```

INVESTIGATION STEPS

1. In real env: decode the Base64 string using CyberChef.
2. Identify the parent process (InitiatingProcessFileName).
3. Check if Defender EDR raised an alert on the device.
4. Review network connections from the device at the same time.

RESPONSE ACTIONS

- Isolate the endpoint via Defender Live Response.
- Collect decoded payload for malware analysis.
- Run a full AV scan on the device.
- Check for persistence: scheduled tasks, registry run keys.

Output from Above KQL Query:

```
1 StormEvents
2 | where StartTime > ago(365d)
3 | where EventType has_any ("Thunderstorm", "Tornado", "Flash Flood")
4 | extend
5   ExtractedCode = extract(@"([A-Z][a-z]+\s[A-Z][a-z]+)", 1, tostring(EpisodeId)),
6   SuspiciousFlag = iff(DeathsDirect == true, "ENCODED_FLAG_SET", "NORMAL")
7 | where SuspiciousFlag == "ENCODED_FLAG_SET"
8 | project StartTime, DeviceName = State, ProcessName = EventType, EpisodeId,
9   SuspiciousFlag,
10  ExtractedCode,
11  Source
12 | order by StartTime desc
13 | take 20
```

Table 1 + Add visual Stats 2026

StartTime	DeviceName	ProcessName	EpisodeId	SuspiciousFlag	ExtractedCode	Source
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No Rows To Show

KQL Detection Query in Sentinel:

DeviceProcessEvents

```
| where TimeGenerated > ago(1h)
| where FileName in~ ("powershell.exe", "pwsh.exe")
| where ProcessCommandLine has_any (
  "-EncodedCommand", "-enc", "-ec",
  "-EncodedC", "-encod"
)
| extend DecodedHint = extract(
  @"(?:-EncodedCommand|-enc|-ec)\s+([A-Za-z0-9+/=]{20,})",
  1, ProcessCommandLine
)
| project
  TimeGenerated, DeviceName, AccountName,
  ProcessCommandLine, DecodedHint,
  InitiatingProcessFileName,
  InitiatingProcessCommandLine
| order by TimeGenerated desc
```